

Sebastian Davalos

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Education

Florida State University

Expected Spring 2029

Bachelor of Science in Computer Engineering

- Relevant Coursework: Data Structures & Algorithms, Digital Logic Design, Discrete Mathematics, Physics I–II, Computer Data Analysis, Calculus I–III

Florida International University

July 2025

Associate of Arts, Dual Enrollment

Experience & Leadership

Undergraduate Research Assistant – UROP

Fall 2026 – Present

Dr. Zhengguang Lu, Department of Physics, Florida State University

- Annotated and curated **400–500 optical microscope images of graphene flakes at 20–50× magnification**, labeling target flakes while distinguishing dust, contamination, and chip edges for object-detection training.
- Prepared training, validation, and test splits in **Roboflow**, reviewing annotation consistency and dataset quality across multiple model iterations.
- Trained and evaluated a **Roboflow RF-DETR Medium** object-detection model, achieving **86.8% mAP@50**, **85.1% recall**, **74.4% precision**, and **79.4% F1** on the validation set.

AWS Cloud Club at ACM at Florida State University

May 2026 – Present

Project Chair

- Conceived and co-lead development of **Matchob**, an AI-assisted resume-tailoring browser extension with **15+ contributors** across frontend, backend/AWS, AI, data, and testing workstreams.
- Coordinate project scope, technical ownership, GitHub issues/epics, milestone planning, onboarding, and cross-team integration.
- Help architect an AWS-native system using **API Gateway**, **Lambda**, **Amazon Bedrock**, **S3**, **IAM**, and **CloudWatch** to generate evidence-grounded resume recommendations while preventing unsupported claims.

CompNeuroSociety at Florida State University

Apr. 2026 – Present

Co-President — Workshop Coordinator, Feb. 2026 – May 2026

- Co-lead a **~40-member** computational neuroscience organization, coordinating executive-board operations, GBMs, funding, and semester programming.
- Awarded a **\$13,000 Open Research Community Accelerator (ORCA) grant** to support training in open science, reproducible research, and computational workflows.
- Coordinate an **8-workshop computational neuroscience series** covering calculus, Python, Brian2, and AI-assisted research, supporting workshop leaders and members as they develop accompanying mini-projects.

Campus Phone Repair

Oct. 2026 – Present

Technician Intern

- Support diagnosis and repair of consumer electronics including smartphones, tablets, and computers, gaining hands-on experience with hardware troubleshooting and device repair.
- Assist with identifying hardware faults, replacing damaged components, and restoring malfunctioning devices under technician supervision.

Projects

StudyStack AI — *Next.js, TypeScript, Supabase, OpenAI*

Feb. 2026 – Apr. 2026

- Developed a full-stack AI study platform for PDF-grounded chat, flashcards, quizzes, and persistent study threads.
- Created a PDF pipeline for extraction, chunking, reprocessing, hybrid retrieval, and source-grounded AI responses.
- Implemented Supabase authentication, private document storage, secure preview/download, and persistent Q&A history to support multi-user study workflows.

SmartGallery — *Python, AWS Rekognition, S3, Git/GitHub*

Jan. 2026 – Apr. 2026

- Built a Python CLI that processes S3-hosted images with AWS Rekognition and returns detected labels and confidence scores.
- Added configurable image inputs and raw JSON exports for faster testing, debugging, and Rekognition response inspection.
- Contributed within a **40-person development team** using GitHub branches, pull requests, code reviews, and shared workflows to integrate cloud-based image recognition.

The Actuary — *Python, Scikit-learn, Pandas*

May 2025 – Oct. 2025

- Made an insurance-risk ML pipeline combining supervised classification, clustering, and anomaly detection.
- Trained a Random Forest model to **98.5% accuracy** using applicant features including age, BMI, and smoking status.
- Applied K-means to group applicants into risk segments and Isolation Forest to flag roughly **5% of records** as anomalous cases for further review.

Technical Skills

Languages: Python, C/C++, Java, SQL, TypeScript

Frameworks/Libraries: React, Next.js, Scikit-learn, Pandas, NumPy

Cloud/Tools: AWS (Lambda, Bedrock, S3, API Gateway), Supabase, Git/GitHub, Linux, Roboflow